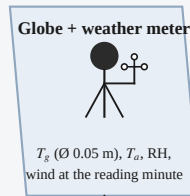
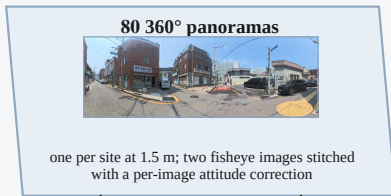


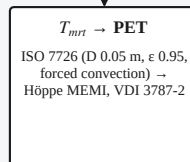
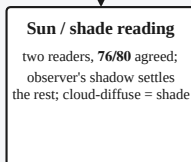
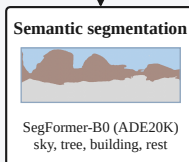
## A Field reference

80 sites · same site, same minute

INPUTS



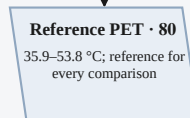
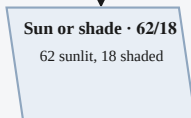
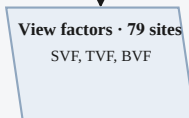
PROCESSING



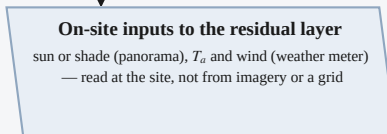
one pose failure  $\rightarrow$  79;  
residual class median 0.02

globe rise 21.7 K sunlit vs  
9.4 K shaded,  $p < 0.001$

OUTPUTS



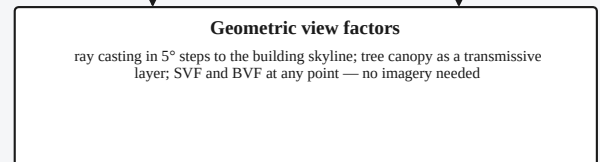
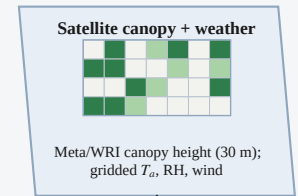
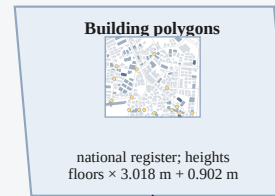
LEARNING



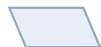
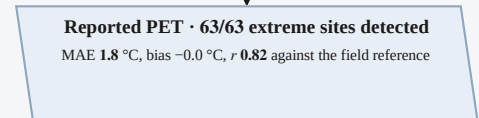
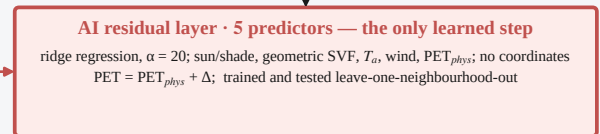
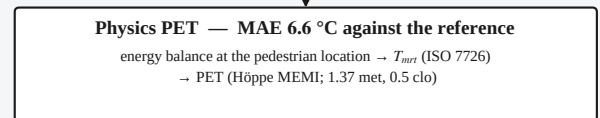
RESULT

## B Imagery-free service

no street view anywhere in the chain



verified against the panorama SVF at the 79 field sites:  
MAE 0.112, bias +0.046,  $r$  0.70



input or output data



processing step



learned step

verification

field data into the service chain